

Product Requirement Document

Product Name

Smart Task Management System

Problem Statement

Teams and individuals struggle with inefficient task organization, leading to missed deadlines, lack of progress visibility, and difficulty in prioritizing critical work. There is a need for a centralized system to streamline task management.

Goals

- Enable users to efficiently create, organize, and manage tasks.
- Facilitate clear task assignment and ownership within teams.
- Provide comprehensive tracking of task progress and deadlines.
- Improve overall team productivity and collaboration through better task visibility.
- Offer intuitive and visual dashboards for task status and workload insights.

User Personas

- Project Manager: Needs to assign tasks, monitor overall project progress, and manage team workload.
- Team Member: Needs to view assigned tasks, update progress, and manage personal tasks.
- Individual Contributor: Needs to organize personal to-do lists and track individual deadlines.
- Team Lead: Needs to oversee team performance, ensure task completion, and identify bottlenecks.

Functional Requirements

- User authentication and authorization (login, signup, password reset).
- Create, edit, and delete tasks with attributes: title, description, due date, priority, assignee(s).
- Assign tasks to one or multiple users.

- Update task status (e.g., To Do, In Progress, Review, Done).
- Set and manage task deadlines with optional reminders.
- Filter and sort tasks by various criteria (assignee, priority, status, due date).
- Search functionality for tasks and projects.
- Dashboard view displaying overall project/team progress, task distribution, and upcoming deadlines.
- Personalized user dashboards showing assigned tasks and individual progress.
- Ability to create and manage projects/groups to categorize tasks.

Non Functional Requirements

- Performance: The system should load pages and perform actions within 2 seconds under normal load.
- Scalability: The system should be able to support up to 10,000 active users and millions of tasks without significant performance degradation.
- Security: All user data and task information must be encrypted in transit and at rest. Robust access controls must be implemented.
- Usability: The user interface must be intuitive, easy to navigate, and require minimal training.
- Reliability: The system should have an uptime of 99.9% annually.
- Compatibility: The web application must be fully functional across modern web browsers (Chrome, Firefox, Edge, Safari).

Constraints

- Initial release must be completed within 6 months.
- Development budget is capped at \$X.
- The system must comply with relevant data privacy regulations (e.g., GDPR, CCPA) for user data handling.
- Reliance on cloud-based infrastructure providers.

Assumptions

- Users have a stable internet connection and access to a modern web browser.

- Users are familiar with basic web application interactions.
- Timely feedback and collaboration will be provided by key stakeholders during the development lifecycle.
- A dedicated and skilled development team is available for the project duration.
- The scope of integrations with other systems will be minimal in the initial phase.

Out Of Scope

- Native mobile applications (Android/iOS) in the initial release.
- Advanced Gantt chart or critical path analysis.
- Built-in chat or real-time messaging features within tasks/projects.
- Integration with third-party calendar applications (e.g., Google Calendar, Outlook Calendar).
- Detailed time tracking per task or billing functionality.
- Resource allocation and workload balancing features beyond basic assignment view.
- Complex reporting tools for sprint velocity or burn-down charts.